

Overview

The **GitHub Release Navigator** (Service 2) is an advanced, autonomous agentic tool designed to navigate GitHub repositories and extract release information without relying on brittle, hardcoded CSS selectors. By leveraging Model Context Protocol (MCP), Computer Vision (GPT-4o), Claude (3.5 Sonnet) and Reinforcement Learning (RL) heuristics, the system emulates human-like browsing behavior to robustly locate and retrieve data.

The Problem

Traditional web scraping relies on static XPaths or CSS selectors, which break immediately when a website updates its UI. Navigating complex, dynamic platforms like GitHub to find specific nested information (e.g., the latest release tag, commit hash, and author) requires a system that can "see" and "understand" the page structure rather than just parsing the DOM blindly.

The Solution

We have built a vision-based navigation agent that decouples the "seeing" (Vision) from the "doing" (Action) and "planning" (Reasoning). The system uses a **Lesser A*** algorithm to identify interactive elements on a page and a **Greater A*** algorithm to score and decide the optimal navigation path based on semantic relevance to the goal.

Key Components

1. MCP Server (`mcp_server.py`)

- Acts as the bridge between the AI agent and the browser (Playwright).
- Exposes granular tools such as `navigate_to_url`, `click_element`, `get_screenshot`, and `get_dom_tree`.
- Manages "Limbs" of the agent, handling browser state, tabs, and direct interactions.

2. Navigator Agent (`navigate.py`)

- **The Controller**: Orchestrates the mission lifecycle.
- **RL-Driven Search**: Implements a two-tiered A* search utilizing a `CostManager` to track "energy" usage (tokens/actions) and a `KnowledgeBase` for Q-learning (memory of successful paths).
- **Prediction Model**: Generates a weighted "bag of words" model specific to the target repository to score links and buttons relevance.

3. Vision & Intelligence (`vision_helper.py`)

- **The Eyes (Tagging Model)**: Uses GPT-4o (Vision) to analyze screenshots, describing

page layout and identifying visual cues (e.g., "Latest" badges).

- **The Brain (Strategy Model):** Uses Claude (or GPT-4o) to reason about the navigation plan, determining if the current state satisfies the goal or if further navigation is required.
- **Hybrid Extraction:** Combines visual analysis with HTML parsing to extract structured data (Version, Commit, Author) with high accuracy.

Capabilities

- **Autonomous Navigation:** Given a repository URL (or just a name), the agent finds its way to the releases page.
- **Self-Correction:** Uses vision to verify if it is on the correct page before attempting extraction.
- **Robustness:** Does not break with minor UI changes, as it looks for semantic and visual meaning (e.g., a green button usually means "Go" or "Primary Action").
- **Real-time Streaming:** Capable of streaming logs, screenshots, and state updates via WebSockets (integrated via `app_wrapper.py`) to external services.

State Representation & Information Architecture

To efficiently map the environment without overwhelming the LLM context, we decomposed the site information into three structured parts:

1. **Formatted DOM Tree:** A cleaned, semantic representation of the page structure (processed via BeautifulSoup).
2. **Navigation Graph:**
 - **Current URL:** The precise location string.
 - **Outbound Links:** A structured list of all HREF links leading out of the current page.
3. **Action Space with Heuristics:** A dynamically generated list of all available actions (clicks, input, navigation) paired with their **heuristic scores**. These scores are calculated locally to prioritize the most promising steps.

Trade-offs & Innovations

Efficient A* Algorithm (Cost Reduction)

One of the most significant achievements of this system is the implementation of a custom **A* Search Algorithm**. Instead of asking the Large Language Model (LLM) "what do I do next?" at every single step—which is slow and expensive—the agent calculates heuristic scores for links and buttons locally.

- **Result:** This drastically saves on API credits and token usage. The agent navigates "autonomously" using these heuristics and only calls the models when it needs a high-level

strategic check or visual confirmation.

Multi-Modal Cognition Strategy

We strategically delegated cognitive tasks to the models best suited for them, creating a robust "Team of Experts" architecture:

- **GPT-4o (The Eyes):** Dedicated to **Vision**. It handles instantaneous visual recognition of page layouts, badges, and "Latest" indicators.
- **Claude 3.5 Sonnet (The Brain):** Dedicated to **Thinking**. It handles complex reasoning, planning, and evaluating if the mission goal is met.
- **A* Algorithm (The Instinct):** Dedicated to **Navigation**. It handles the rapid, low-cost pathfinding based on text patterns and DOM structure without model inference.

Trade-off: Latency vs. Robustness

While this agentic approach introduces higher latency compared to a hardcoded script (due to the "thinking" time), the trade-off yields significantly higher **robustness**. The system does not break when CSS classes change, ensuring long-term reliability and reduced maintenance overhead.

Conclusion

This system represents a shift from fragile scripting to **resilient, agentic automation**. It successfully solves the problem of extracting structured data from dynamic environments by mimicking human visual processing and decision-making logic.

Future Directions

The system has been successfully validated against the `openclaw` repository and edge cases (non-existent repos). Moving forward, the roadmap focuses on user accessibility and autonomy:

1. **Frontend Interface:** Developing a visual dashboard to monitor the agent's real-time choices and visual stream.
2. **Autonomous CLI Tool:** Packaging the solution into a standalone, easy-to-install CLI for widespread developer use.

Ultimate Vision: To become the *"OpenClaw for OpenClaw"*—a meta-tool that autonomously navigates, understands, and interacts with the open-source ecosystem as effectively as a human contributor.